

Caelum Thornton

Contact Information |

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Professional Summary

Highly motivated and results-oriented Machine Learning Research Fellow at Quantum Horizon Corp. with a proven track record in developing and deploying innovative AI/ML solutions. Possessing a strong theoretical foundation from MIT and practical experience in applying cutting-edge techniques to solve complex real-world problems. Expertise in Python, Java, Go, TensorFlow, Keras, and GCP, with a demonstrated ability to collaborate effectively within multidisciplinary teams. Seeking a challenging role where I can leverage my skills and experience to contribute to the advancement of AI and its applications.

Education

Massachusetts Institute of Technology (MIT), Cambridge, MA |

Master of Science in Computer Science, Artificial Intelligence Focus | May 2021 | GPA: 3.9/4.0 | Thesis: "Novel Applications of Reinforcement Learning in Autonomous Navigation"

Massachusetts Institute of Technology (MIT), Cambridge, MA |

Bachelor of Science in Computer Science | May 2019 | GPA: 3.8/4.0 | Minor: Mathematics

Technical Skills

Programming Languages: Python (Expert), Java (Proficient), Go (Intermediate), C++ (Basic) **Machine Learning Frameworks:** TensorFlow (Expert), Keras (Expert), PyTorch (Proficient), Scikit-learn (Proficient) **Cloud Platforms:** Google Cloud Platform (GCP) (Expert), Amazon Web Services (AWS) (Intermediate) **Databases:** SQL (Proficient), NoSQL (Intermediate) **Big Data Technologies:** Spark (Intermediate), Hadoop (Basic) **Other:** Git, Docker, Kubernetes, Agile methodologies

Professional Experience

Machine Learning Research Fellow | Quantum Horizon Corp., Boston, MA | June 2021 – Present

- * Developed and implemented a novel deep learning model for anomaly detection in high-frequency financial data, resulting in a 15% improvement in accuracy compared to existing methods.
- * Led a team of three junior engineers in the design and deployment of a real-time fraud detection system using TensorFlow and GCP.
- * Contributed to the development of a proprietary reinforcement learning algorithm for optimizing resource allocation in cloud infrastructure.
- * Published two research papers on the applications of AI in finance and presented findings at international conferences.
- * Mentored and trained junior researchers in advanced machine learning techniques.

Software Engineering Intern | InnovateTech Solutions, Cambridge, MA | June 2020 – August 2020

- * Developed and maintained several key features for a large-scale data processing pipeline using Java and Spark.
- * Implemented a new data visualization module that improved user experience and data analysis efficiency.
- * Contributed to the improvement of software development processes through the implementation of agile methodologies.

AI/ML Projects

Anomaly Detection in Financial Time Series Data |

- * Developed a deep learning model using LSTM networks and autoencoders to detect anomalies in high-frequency financial data.
- * Achieved a 15% improvement in accuracy compared to traditional methods.
- * Deployed the model on GCP using Kubernetes for scalability and reliability.

Real-time Fraud Detection System |

- * Designed and implemented a real-time fraud detection system using TensorFlow and GCP.
- * Achieved a 98% accuracy rate in detecting fraudulent transactions.
- * The system is currently being used by a major financial institution.

Reinforcement Learning for Resource Allocation |

- * Developed a novel reinforcement learning algorithm for optimizing resource allocation in cloud infrastructure.
- * The algorithm resulted in a 10% reduction in resource consumption.
- * The algorithm is currently being patented.

Image Recognition System for Medical Diagnosis |

- * Developed a convolutional neural network (CNN) for the automatic diagnosis of medical images.
- * Achieved a 95% accuracy rate in classifying images.
- * The system is currently undergoing clinical trials.